APEX

Formula 1 Race Performance Analysis

Project Documentation Report

Project by	**Mehvish Siddiqui**
Project type	Data analysis, visualisation, storytelling, and web integration
Tools used	Tableau Public, HTML, CSS, GitHub, GitHub Pages
Submission components	GitHub repository, live website, recorded walkthrough

MEDICAPS UNIVERSITY, INDORE

B.Tech. Computer Science and Engineering (Data Science)

Academic Session: 2026–2027

Year/Semester: Second Year | Semester III

Live website: https://mehvish04.github.io/f1-race-performance/

GitHub repository: https://github.com/mehvish04/f1-race-performance

Recorded walkthrough: https://drive.google.com/file/d/1XBUo5rdXJ7oOkrz5R8-M09nxMysB7P_z/view?usp=sharing

1. Abstract

This project presents a visual analysis of Formula 1 race performance using Tableau Public and a custom-built website. The analysis focuses on race wins across seasons, driver performance based on total points, and team performance based on total points. The Tableau worksheets are organised into a dashboard and a story, then embedded in a website named APEX using HTML and CSS. The website is hosted through GitHub Pages, while the source files are maintained in a GitHub repository.

2. Objectives

- Explore Formula 1 race data and prepare it for visual analysis.
- Create visualisations that communicate race wins and driver/team point comparisons.
- Combine the visualisations into an accessible dashboard.
- Create a Tableau story that presents the analysis in a sequence of focused views.
- Build a branded web page to present the published Tableau dashboard and story.
- Publish the source code and live website, and demonstrate the completed workflow in a recorded video.

3. Tools and Technologies

Tool / technology	Purpose
Microsoft Excel workbook	Provided source data containing Formula 1 race-related tables/sheets.
Tableau Public	Data exploration, calculated field, worksheets, dashboard, story, and publishing.
HTML	Structure of the APEX project website.
CSS	Website layout, typography, colours, spacing, and visual styling.
GitHub	Versioned hosting of the website source files.
GitHub Pages	Publication of the static website on the web.
macOS screen recording	Recording the project demonstration with voice narration.

4. Dataset Overview

The project uses a university-provided Excel workbook. The workbook includes sheets named race_results, championship_standings, circuits, qualifying_results, and lap_intelligence. The primary worksheet used for the reported visualisations was race_results, which contained approximately 6,464 rows and 42 fields in the working dataset.

The analysis described in this report is based on the selected race-results data and the visualisations created from it. The report does not claim that every sheet in the workbook was incorporated into the final dashboard.

5. Data Preparation and Calculated Field

The race_results sheet was used as the primary source for the three core visualisations. A calculated field named Race Wins was created in Tableau to identify race wins from the finish-position field.

```
IF [Finish Position] = 1 THEN 1 ELSE 0 END
```

This calculation assigns a value of 1 when Finish Position equals 1 and 0 otherwise. Aggregating this field makes it possible to count wins within the relevant season or category. Before final submission, confirm that null, non-numeric, or special finish-position values are handled appropriately for the dataset and the intended analysis.

6. Visualisation Development

6.1 Race Wins by Season

This worksheet presents race wins grouped by season. It is intended to help viewers compare winning totals across seasons and observe changes over time.

6.2 Driver Performance by Total Points

This worksheet compares drivers using total points, with a Top 10 driver filter applied to focus the view on a smaller set of leading entries.

6.3 Team Performance by Total Points

This worksheet compares teams using total points, allowing viewers to examine relative team performance in the selected data.

7. Dashboard Design

The three worksheets were combined into a Tableau dashboard titled “Formula 1 Race Performance Analysis.” The dashboard places the visualisations in a vertical layout so the viewer can move from seasonal race wins to driver and team comparisons. The dashboard was configured with a fixed size of approximately 1250 × 1400 pixels. Depending on the browser window and embedded frame dimensions, scrolling may be needed to view all content.

8. Tableau Story

A Tableau story was created with four story points: the overall dashboard, Race Wins by Season, Driver Performance by Total Points, and Team Performance by Total Points. This structure gives the audience an overview first, followed by focused views of each main analytical question.

9. Website Development and Integration

1. Create the APEX page using HTML for page structure.

2. Use CSS to apply the project's midnight-navy and gold visual identity, typography, spacing, and layout.
3. Publish the dashboard and story to Tableau Public.
4. Embed the published Tableau views in the website using embedded frames.
5. Save the website source file(s), including index.html, in the GitHub repository.
6. Configure GitHub Pages to publish the static website from the main branch and repository root.
7. Open the live website and check that the page and embedded Tableau views load.

Live website: https://mehvish04.github.io/f1-race-performance/

GitHub repository: https://github.com/mehvish04/f1-race-performance

Tableau Public dashboard: https://public.tableau.com/app/profile/mehvish.siddiqui/viz/Formula1RacePerformanceAnalysis/Dashboard1?publish=yes

Tableau Public story: https://public.tableau.com/app/profile/mehvish.siddiqui/viz/Formula1RacePerformanceAnalysis/Story1?publish=yes

10. Testing and Validation

The following functional checks were performed during project development: the published website was opened; the embedded dashboard and story were checked for display; basic story navigation was tested; chart tooltips and dashboard interaction were checked. The dashboard may require scrolling because of its fixed layout.

Formal performance benchmarking has not been documented in this report. Since performance testing is an important part of the assessment rubric, additional measurements should be collected before submission if required. Recommended checks include:

- Record the website's initial load time across several repeat runs and report the method and results.
- Measure or document the time taken for the embedded Tableau views to become usable.
- Test the page at a desktop size and a narrower browser width; note clipping, scrolling, or layout changes.
- Check that the website and embeds remain accessible after a page refresh.
- Capture screenshots of the test conditions and results; do not report estimated values as measured results.

11. Project Demonstration Video

A screen-recorded walkthrough with voice narration was created to demonstrate the website and project components. The walkthrough includes the APEX website, dashboard visualisations, Tableau story, and source repository. The final video should be reviewed before submission to ensure that narration is audible, the relevant visuals are visible, and blank or loading intervals are minimised.

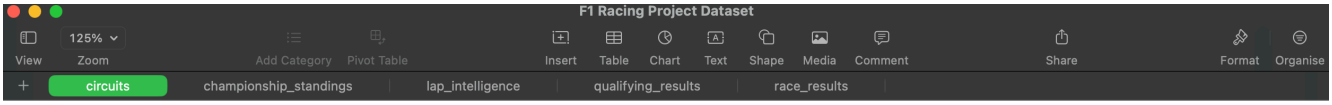
Google Drive video URL: https://drive.google.com/file/d/1XBUo5rdXJ7oOkrz5R8-M09nxMysB7P_z/view?usp=sharing

12. Conclusion

The Formula 1 Race Performance Analysis project combines data visualisation, dashboard design, analytical storytelling, and web integration. Tableau Public was used to create three performance visualisations, combine them into a dashboard, and organise them into a story. A custom HTML/CSS website named APEX presents the published views, while GitHub and GitHub Pages provide source-code hosting and public website deployment. The project demonstrates an end-to-end workflow from data exploration to a shareable web presentation. Further measured performance testing and supporting screenshots can improve the completeness of the documentation.

Appendix A. Screenshot Checklist

- Source workbook and relevant sheets



circuit_id	circuit_name	country	city	laps	lap_km	type	altitude_m	direction
bahrain	Bahrain International Circuit	Bahrain	Sakhir	57	5.412	Permanent	7	Clockwise
jeddah	Jeddah Corniche Circuit	Saudi Arabia	Jeddah	50	6.174	Street	5	Clockwise
albert_park	Albert Park Circuit	Australia	Melbourne	58	5.278	Street/Park	10	Clockwise
suzuka	Suzuka International Circuit	Japan	Suzuka	53	5.807	Permanent	42	Mixed
shanghai	Shanghai International Circuit	China	Shanghai	56	5.451	Permanent	5	Clockwise
miami	Miami International Autodrome	USA	Miami	57	5.412	Street/Park	2	Clockwise
imola	Autodromo Enzo e Dino Ferrari	Italy	Imola	63	4.909	Permanent	38	Anti-clockwise
monaco	Circuit de Monaco	Monaco	Monte Carlo	78	3.337	Street	7	Clockwise
canada	Circuit Gilles Villeneuve	Canada	Montreal	70	4.361	Street/Park	8	Clockwise
spain	Circuit de Barcelona-Catalunya	Spain	Barcelona	66	4.657	Permanent	115	Clockwise
austria	Red Bull Ring	Austria	Spielberg	71	4.318	Permanent	678	Clockwise
silverstone	Silverstone Circuit	UK	Silverstone	52	5.891	Permanent	153	Clockwise
hungary	Hungaroring	Hungary	Budapest	70	4.381	Permanent	264	Clockwise
spa	Circuit de Spa-Francorchamps	Belgium	Spa	44	7.004	Permanent	401	Clockwise
zandvoort	Circuit Zandvoort	Netherlands	Zandvoort	72	4.259	Permanent	10	Clockwise
monza	Autodromo Nazionale Monza	Italy	Monza	53	5.793	Permanent	162	Clockwise
baku	Baku City Circuit	Azerbaijan	Baku	51	6.003	Street	28	Anti-clockwise
singapore	Marina Bay Street Circuit	Singapore	Singapore	62	5.063	Street	15	Anti-clockwise
cota	Circuit of the Americas	USA	Austin	56	5.513	Permanent	165	Anti-clockwise
mexico	Autodromo Hermanos Rodriguez	Mexico	Mexico City	71	4.304	Permanent	2240	Clockwise
brazil	Autodromo Jose Carlos Pace	Brazil	São Paulo	71	4.309	Permanent	785	Anti-clockwise
las_vegas	Las Vegas Strip Circuit	USA	Las Vegas	50	6.12	Street	620	Anti-clockwise
qatar	Losail International Circuit	Qatar	Lusail	57	5.38	Permanent	11	Clockwise
abu_dhabi	Yas Marina Circuit	UAE	Abu Dhabi	58	5.281	Permanent	3	Anti-clockwise
istanbul	Istanbul Park	Turkey	Istanbul	58	5.338	Permanent	130	Anti-clockwise
portimao	Autodromo Internacional do Algarve	Portugal	Portimão	66	4.653	Permanent	108	Clockwise
nurburgring	Nürburgring	Germany	Nürburg	60	5.148	Permanent	473	Clockwise
mugello	Mugello Circuit	Italy	Scarperia	59	5.245	Permanent	255	Clockwise
bahrain_outer	Bahrain International (Outer)	Bahrain	Sakhir	87	3.543	Permanent	7	Clockwise

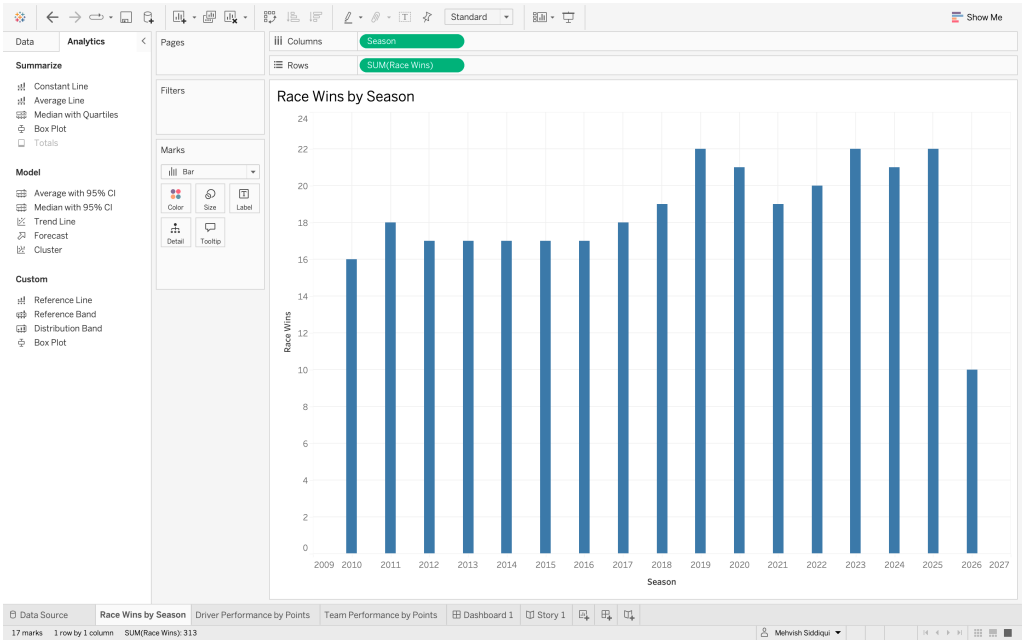
- Tableau calculated field: Race Wins

Race Wins

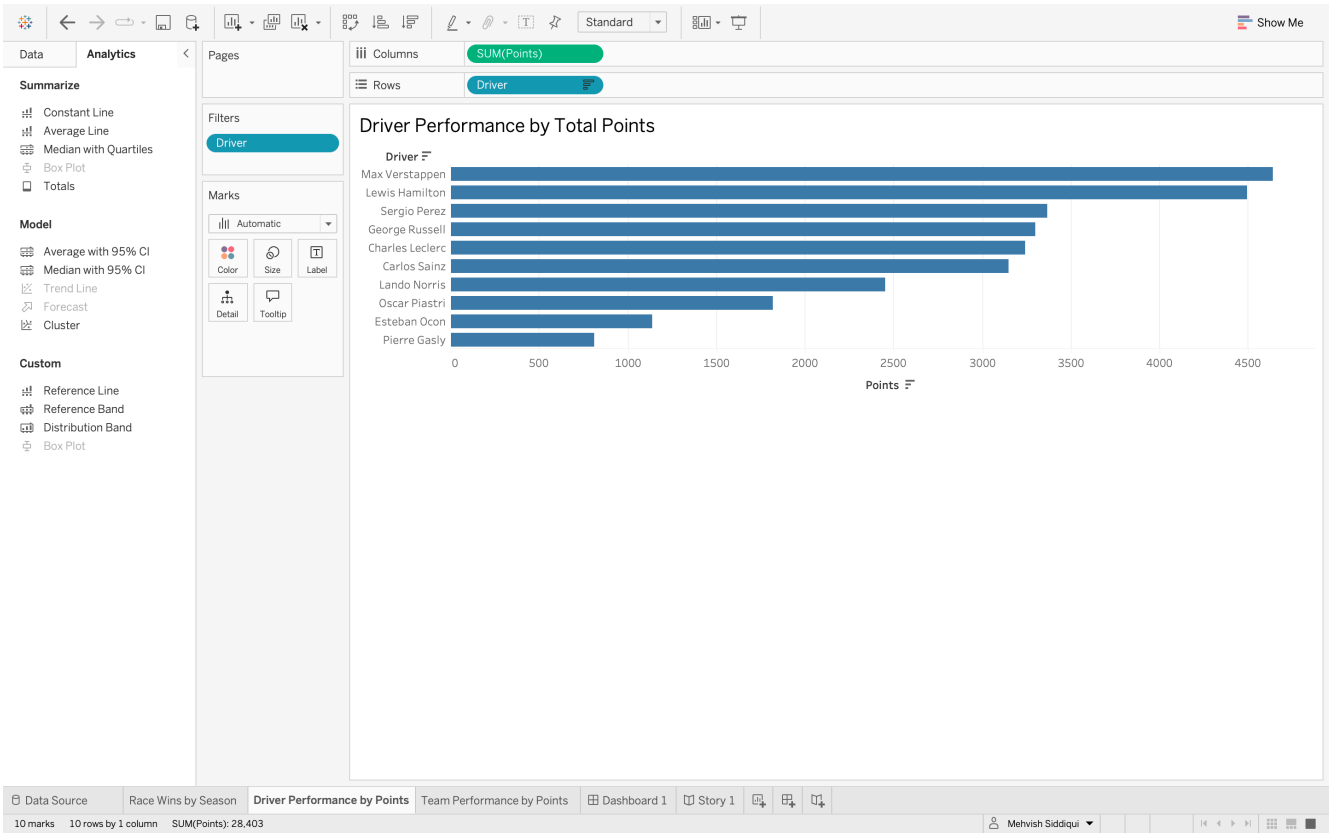
```
IF [Finish Position] = 1 THEN 1
ELSE 0
END
```

The calculation is valid. 3 Dependencies Apply OK

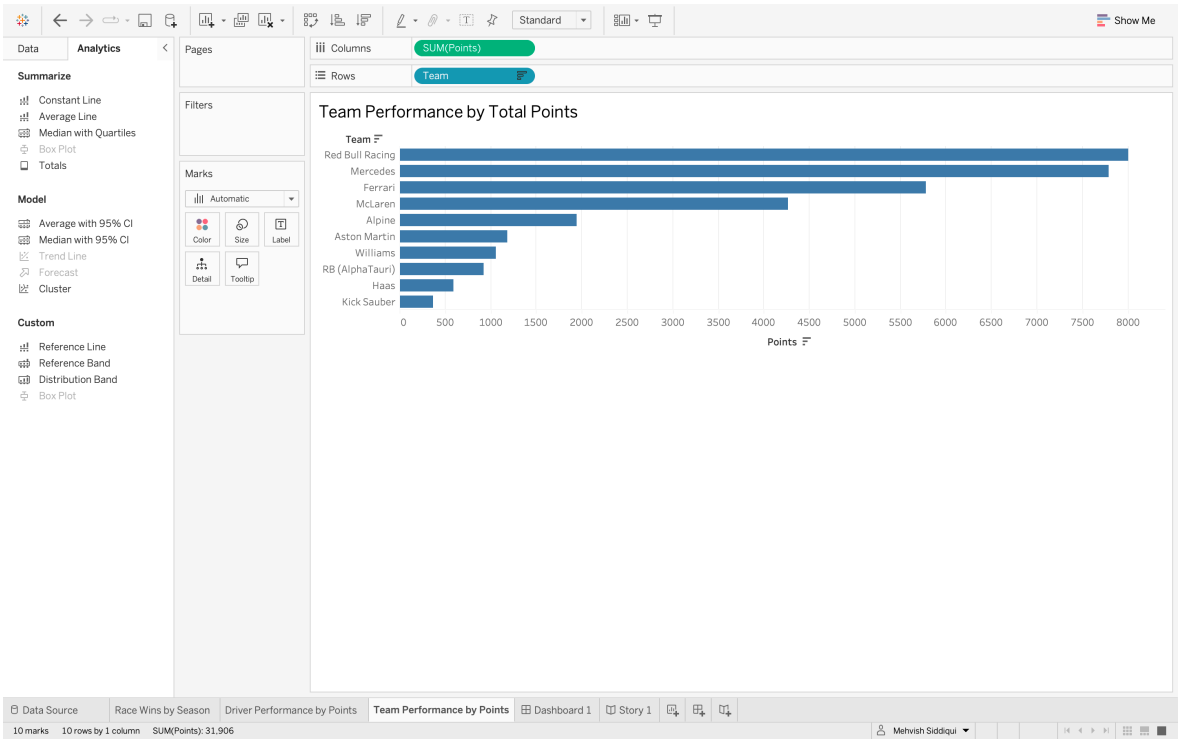
- Race Wins by Season worksheet



- Driver Performance by Total Points worksheet

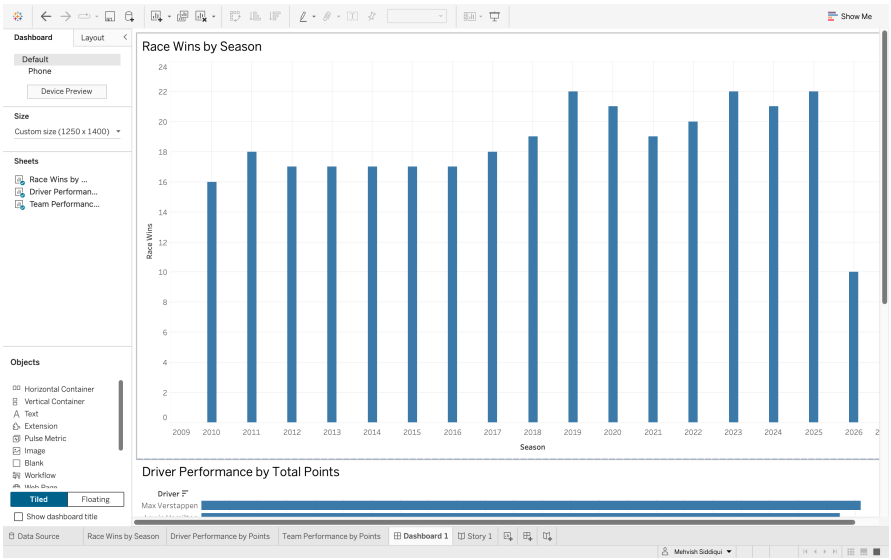


- Team Performance by Total Points worksheet



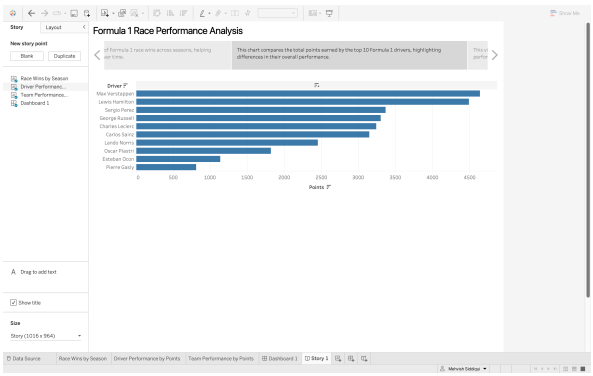
- Complete dashboard

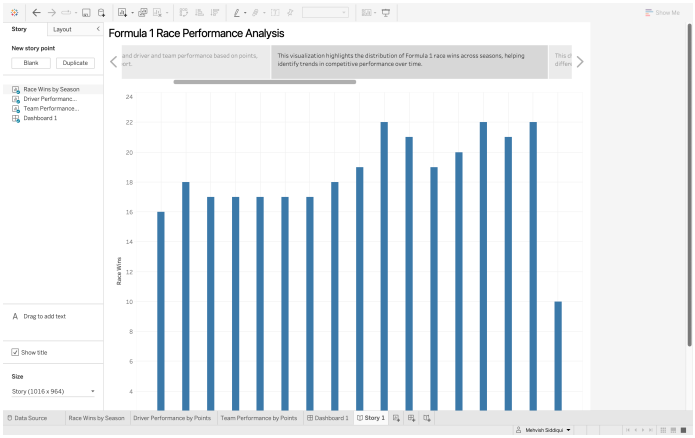


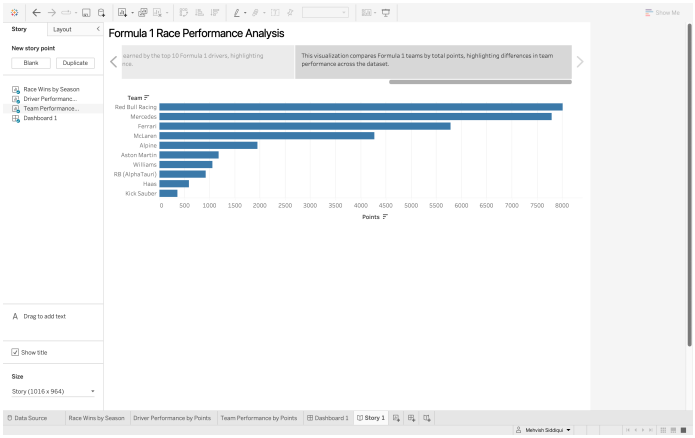


- Four Tableau story points

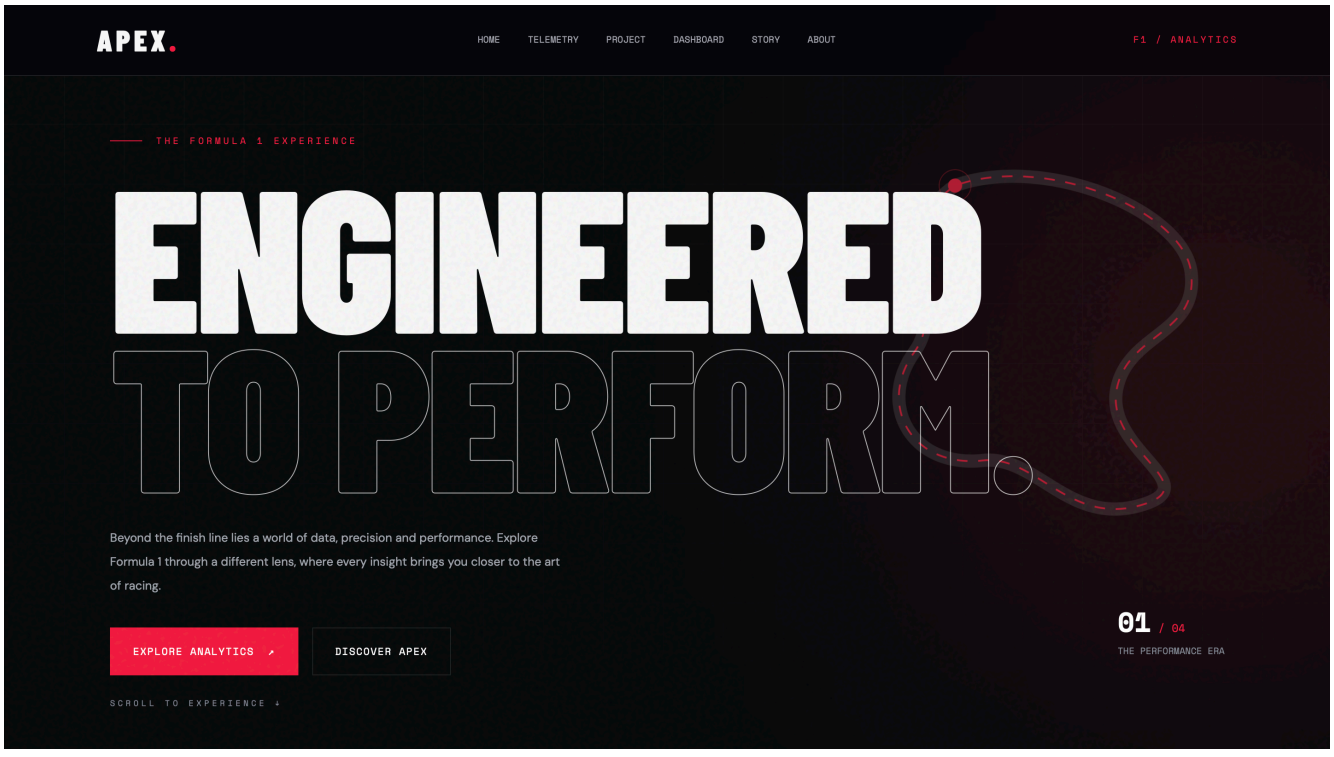




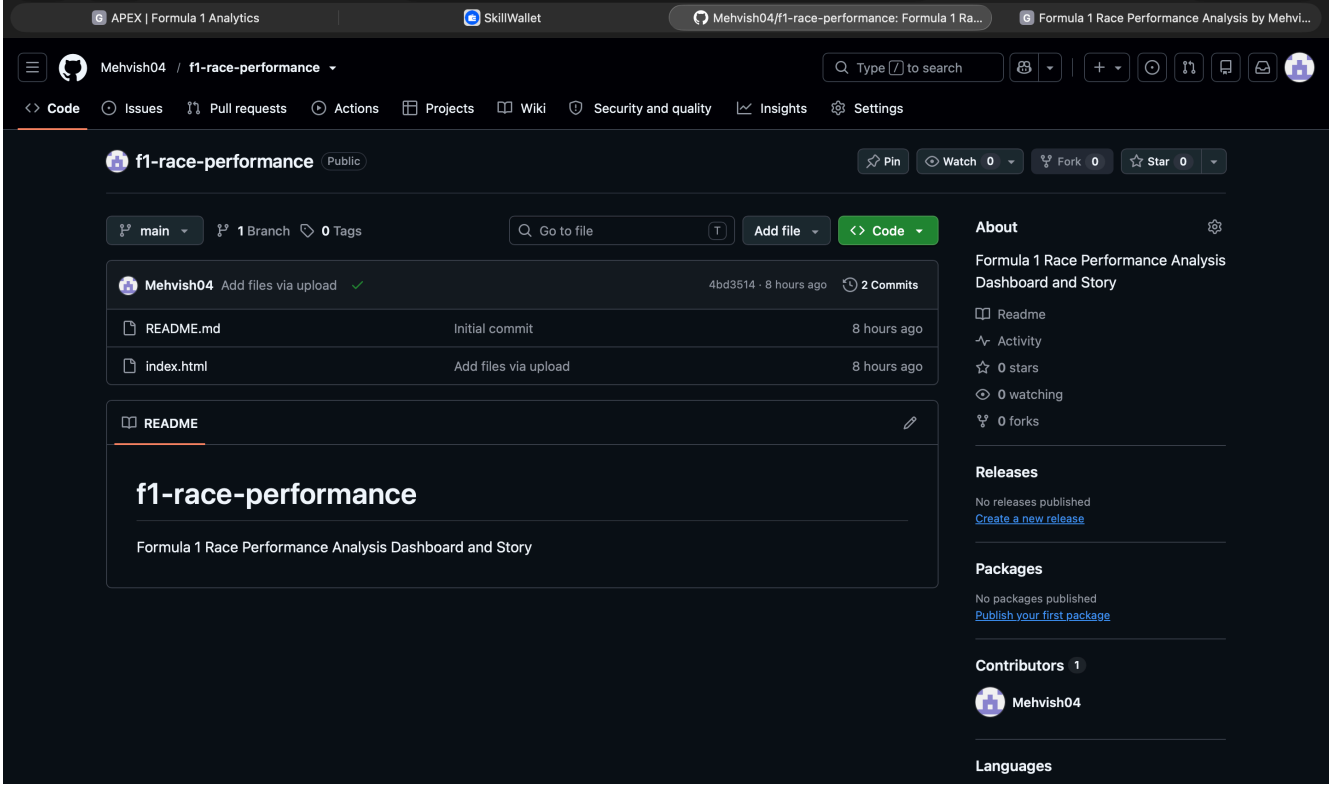




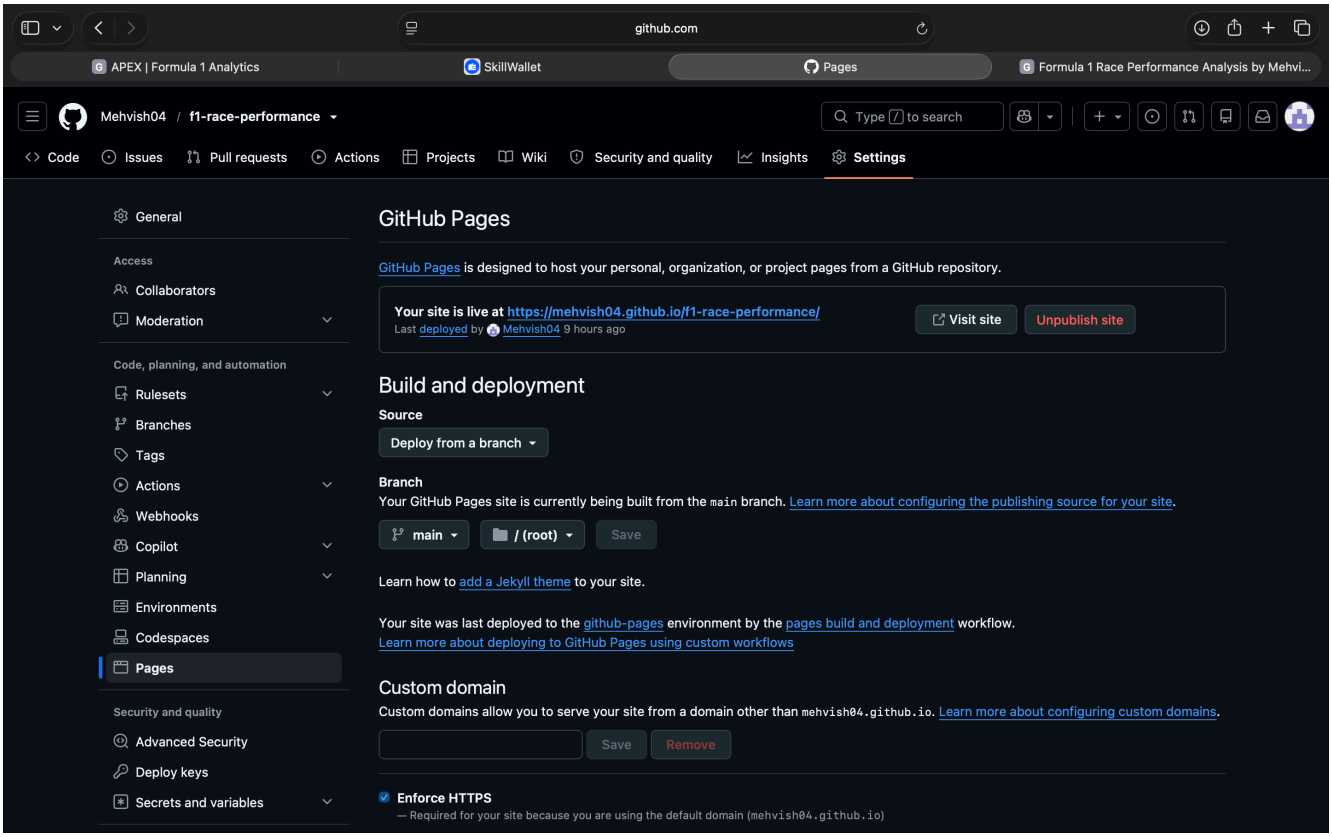
- APEX website



- GitHub repository file list



- GitHub Pages deployment/settings confirmation



- Video Demonstration Evidence

